

Charge them up

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There is a growing understanding in South Africa that electricity prices will have to rise to fund investment in new generating capacity. But how high should prices rise and what would be an efficient price level?

Large South African consumers currently enjoy among the world's lowest prices for electricity. The government views this as good for industry and a means for attracting energy-intensive investments. Part of the reason for the low electricity price is very cheap coal that cannot be profitably exported because of its location and quality.

But the larger part of the reason for the low electricity price is that Eskom's power stations were built over 20 years ago when their cost in money of that day was much lower, financed out of debt now largely repaid, and whose book value is a tiny fraction of their current replacement value. Eskom's 2007 balance sheet shows a book value for its power stations just under R 30 billion - about R700 per kilowatt of installed capacity, or \$US 100/kW. A new coal-fired power station will cost more than \$US 1,500/kW.

Most existing power stations have more than 20 years of remaining life, so their *economic* value is much higher than their book value. If Eskom were required to sell electricity at a price that covered the real value of its assets, rather than their excessively depreciated book value, then prices would more than double. Electricity from a new coal fired power station would cost close to 30c/kWh, whereas Eskom's reported average generation cost is only about 12c/kWh. Furthermore, Eskom is adding 50 per cent more capacity in the next ten years and the value of its assets will multiply. Prices, in any event, will need to rise sharply to fund this new investment. If the Government offers a development tariff at or below the 2007 Eskom average industrial price, then it is effectively providing a massive subsidy to new industries that would be paying generation costs less than one-third of Eskom's extra costs of supply.

One of the consequences of not charging large industrial users the right economic price for electricity is that energy-intensive industries, such as aluminium producers, have chosen to locate in South Africa rather than where electricity costs are truly low. This situation is unlikely to continue. Electricity from new base load coal stations in South Africa cannot compete with surplus, and low-cost, hydro or gas resources in countries such as Iceland or those in the Gulf. And South Africa's coal-fired power stations generate additional costs to the planet in terms of high greenhouse gas emissions.

Eskom is clearly undercharging for its existing supply. If Eskom and the Government were behaving in a commercial way, the regulator would have to set a wholesale price of electricity high enough to reflect the economic value of its assets and to fund new investment.

The usual objections to raising electricity prices are that it would hit the poor and vulnerable, it would increase inflation, and deter foreign investment and development. All these arguments are bogus. If the poor receive a certain number of units each month for free, and prices were doubled, they would still receive these units at no higher cost – twice nothing is still nothing. Nor does raising the industrial price necessarily mean raising domestic prices.

If electricity prices rise to large users and import duties or other purchase taxes fall, then there is no impact on the price level. More important, if the Government acts as a sound shareholder in its companies, requiring a normal (but not excessive) rate of return, and its income goes up, it can reduce borrowing or taxes and reduce inflationary pressures. It could increase investments in other infrastructure sectors and lower prices in telecoms and ports (which, by international standards, are currently very expensive).

South African electricity prices are presently so cheap, that even large price hikes would not affect the competitiveness of existing South African industry. More economically efficient pricing would also lead to more efficient electricity consumption patterns and a less energy and capital intensive, more labour-absorbing, development path.

There is now a growing acceptance that electricity prices in South Africa will have to rise to fund Eskom's new capacity expansion programme. But there is an additional argument that needs to be addressed: the importance of raising electricity prices further to economic levels that will encourage allocative efficiency. Government and the National Energy Regulator will soon face these critical choices with not insignificant consequences for the nature and pace of economic development in South Africa.

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